



Liferay, Alfresco, SSO and LDAP – Full Integration

A generic solution of SSO plus LDAP against
- Liferay Portals and Alfresco portlets
- Alfresco (compliant) standalone applications

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July 18, 2007

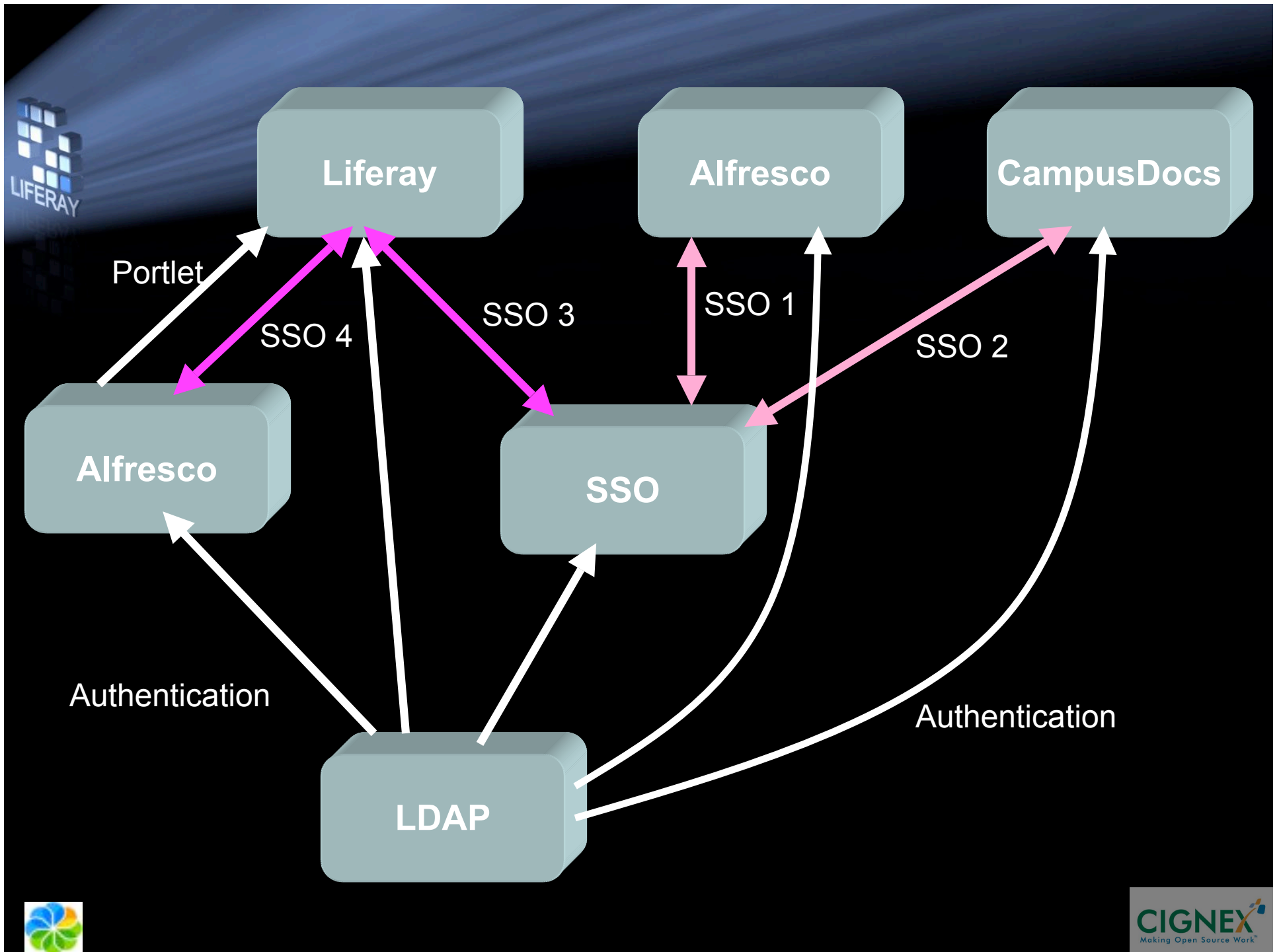




Agenda

1. Overview
2. LDAP & SSO
3. Filters in Action
4. Portlets In Action
5. Real Examples
6. Summary





What is LDAP?

LDAP = Lightweight Directory Access Protocol

- An application protocol for querying and modifying directory services running over TCP/IP.
- A Directory Information Tree (DIT) is data represented in a hierarchical tree-like structure consisting of the Distinguished names (DNs) of the directory entries.



What is SSO?

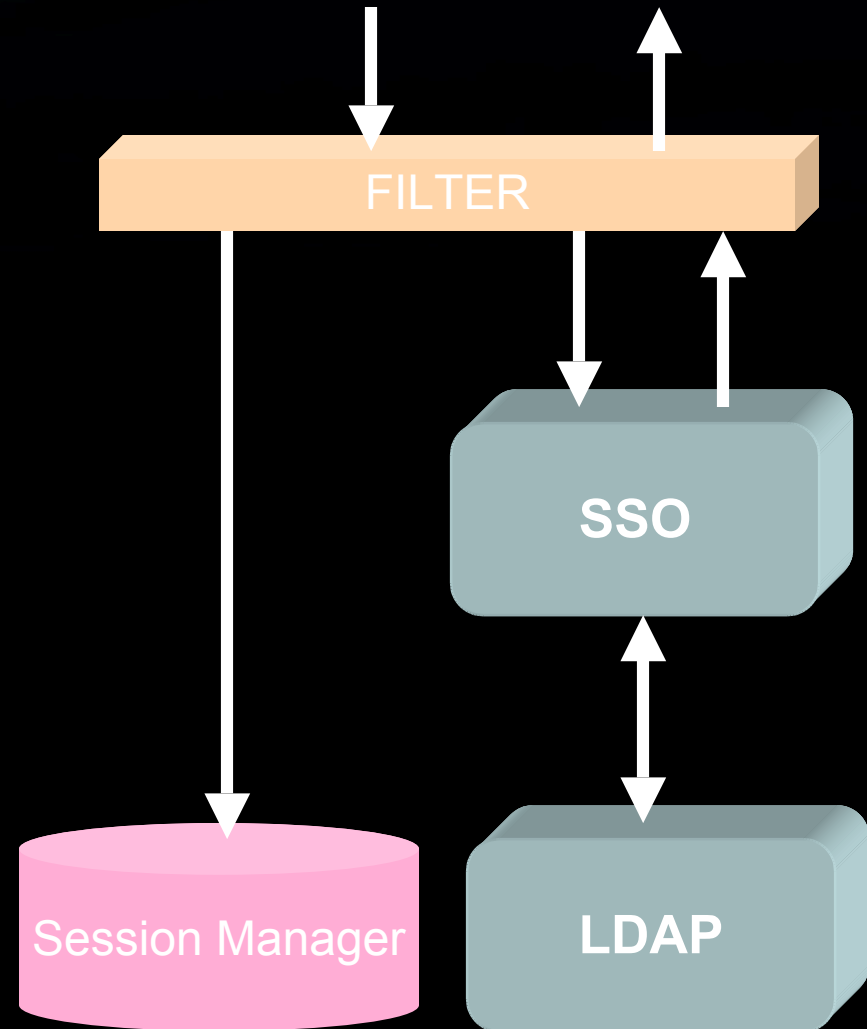
SSO = Single Sign-On

- **Single sign-on (SSO)** is a method of access control that enables a user to authenticate once and gain access to the resources of multiple software systems.
- The JA-SIG Central Authentication Service (**CAS**) is an open single sign-on service that allows web applications the ability to defer all authentication to a trusted central server or servers.



Filters In Action

- Enhanced SSO Filter
- Authenticating SSO Server with LDAP
- Enhanced validating URL in SSO Server
- Authenticating with LDAP
- Log-in Automatically
- E. g. SSO 1, 2, 3





Enhanced SSO Filter

- `<filter><filter-name>CAS Filter</filter-name>`
- `<filter-class>com.cignex.filters.sso.cas.CASFilter</filter-class>`
- `<init-param>`
- `<param-name>cas_server_url</param-name>`
- `<param-value>https://docs.cignex.com/cas-web</param-value>`
- `</init-param>`
- `<init-param>`
- `<param-name>service_path</param-name>`
- `<param-value>/index.jsp</param-value>`
- `</init-param>`
- `<init-param>`
- `<param-name>application_type</param-name>`
- `<param-value>alfresco</param-value>`
- `</init-param></filter>`
- `<filter-mapping><filter-name>CAS Filter</filter-name>`
- `<url-pattern>/index.jsp</url-pattern></filter-mapping>`
- `<filter-mapping><filter-name>CAS Filter</filter-name>`
- `<url-pattern>/logout</url-pattern></filter-mapping>`





Authenticating SSO with LDAP

- Add Adaptors - LDAP Authentication
- Configure LDAP server with LDAP server
- ```
<bean id="contextSource"
 class="org.jasig.cas.adaptors.ldap.util.AuthenticatedLdapContextSource">
```
- ```
  <property name="urls">
```
- ```
 <list>
```
- ```
      <value>ldap://docs.cignex.com:10389</value>
```
- ```
 </list>
```
- ```
  </property>
```
- ```
</bean>
```







# Enhanced validating URL in SSO Server

```
• public static String retrieve(String url) throws IOException {
• ...
• BufferedReader r = null;
• try {
• url = url.replace("https://", "http://"); //add
• URL u = new URL(url);
• URLConnection uc = u.openConnection();
• uc.setRequestProperty("Connection", "close");
• r = new BufferedReader(new InputStreamReader(uc.getInputStream()));
• String line;
• StringBuffer buf = new StringBuffer();
• while ((line = r.readLine()) != null)
• buf.append(line + "\n");
• return buf.toString();
• } finally {
• ... }
• }
```





# Authenticating with LDAP

- **Configurable**

- ```
<bean id="ldapExtInitialDirContextFactory"  
class="com.cignex.portal.security.authentication.ldap.LDAPInitialDirContextFactoryImpl">
```
- ```
 <property name="initialDirContextEnvironment">
```
- ```
    <map>
```
- ```
 <!-- The LDAP provider -->
```
- ```
      <entry key="java.naming.factory.initial">
```
- ```
 <value>com.sun.jndi.ldap.LdapCtxFactory</value>
```
- ```
      </entry>
```
- ```
 <entry key="java.naming.provider.url">
```
- ```
        <value>ldap://docs.cignex.com:10389</value>
```
- ```
 </entry>
```
- ```
      <entry key="java.naming.security.authentication">
```
- ```
 <value>simple</value>
```
- ```
      </entry>
```
- ```
 ...
```
- ```
    </map>
```
- ```
 </property>
```
- ```
</bean>
```





Log-in Automatically

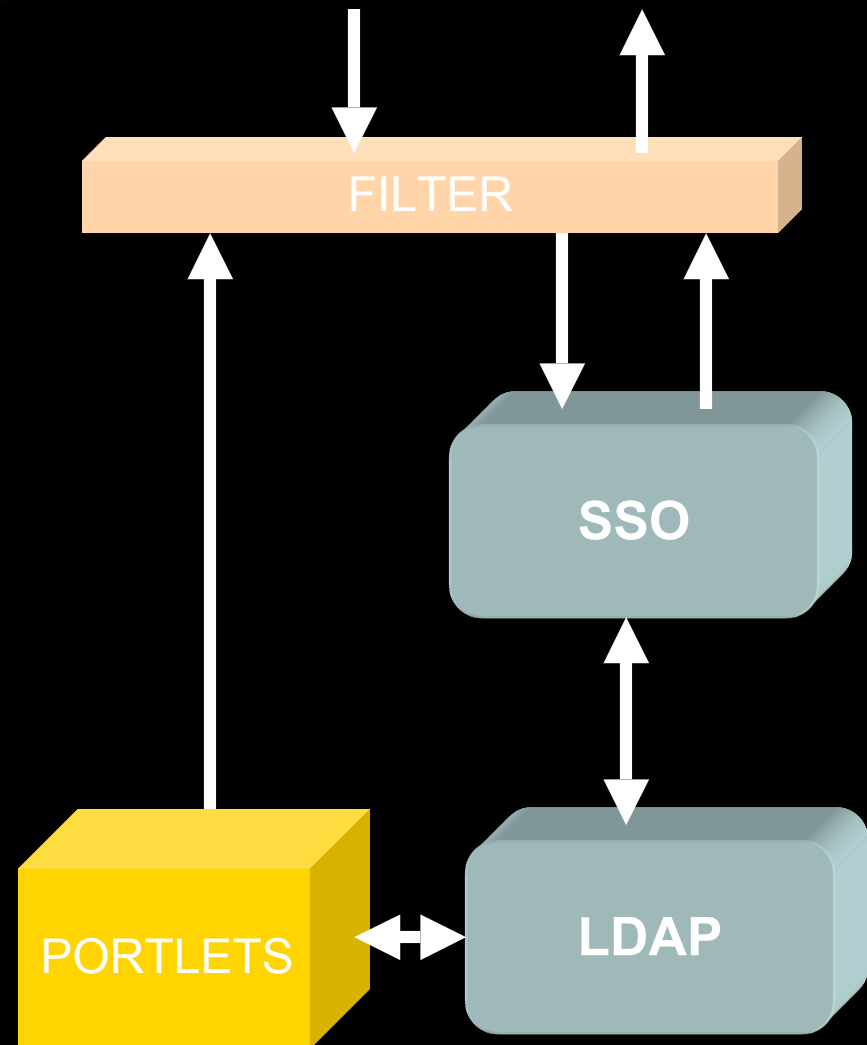
- Reuse log-in mechanism of different applications – Liferay Portal, Alfresco, CampusDocs.
- One Interface, many implementations
- Initialize applications
- Redirection



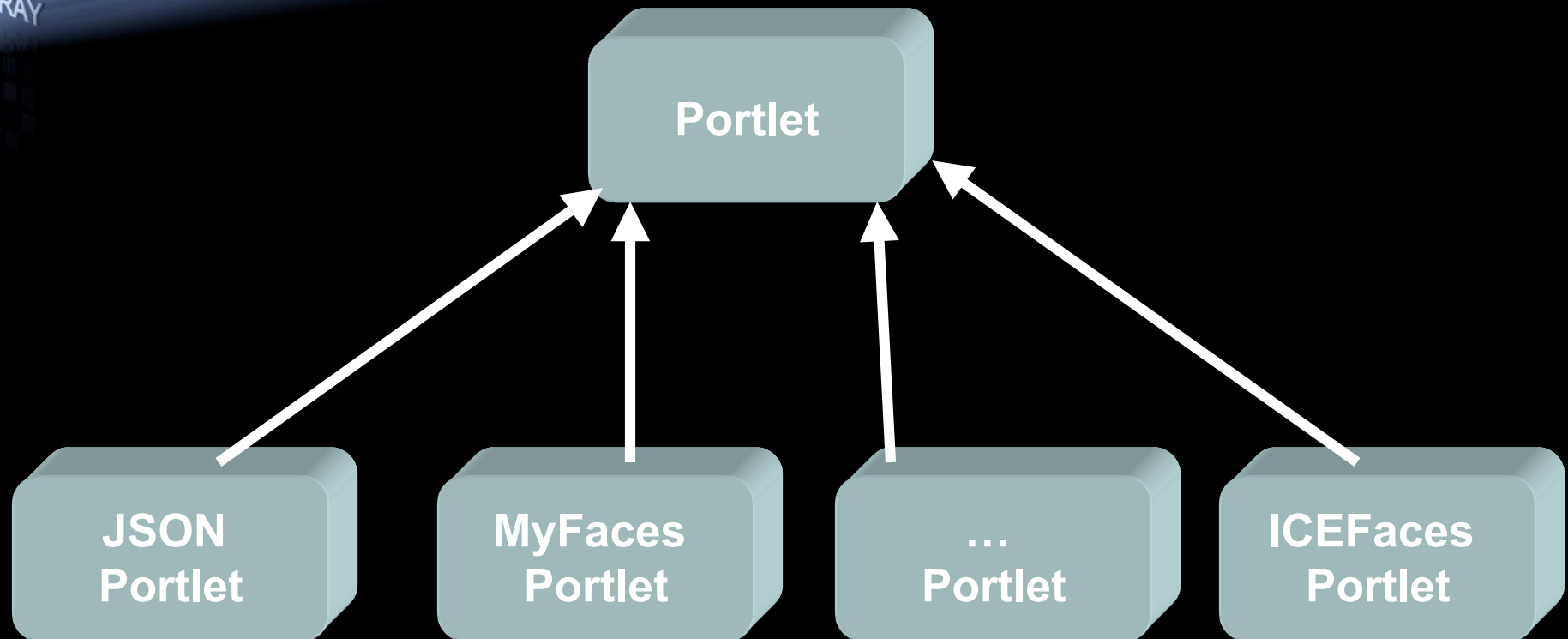


Portlets In Action

- Loading portlet
- Tracing current user from Filter
- Authenticating portlet with LDAP
- Log-in Automatically
- E.g. SSO 4



Generic Portlet Interface





Authenticating Portlet with LDAP

- **Configurable & Reusable**

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- `<!-- Single Sign On Authentication -->`

- `<bean name="ssoExtAuthenticationContext"`
`class="com.cignex.portal.security.authentication.sso.SSOAuthenticationCon`
`text" >`

- `<property name="allowSingleSignOn">`

- `<value>true</value>`

- `</property>`

- `<property name="filter">`

- `<value>CAS_SSO</value>`

- `</property>`

- `</bean>`





Log-in Automatically

- Reuse log-in mechanism of different portlets – Alfresco (MyFaces), CampusDocs (ICEfaces).
- One Interface, multiple implementations
- Initialize portlet
- Trace session & context – Portlet & Servlet
- Redirection





Real Examples

1. SSO CAS plus LDAP ADS against Alfresco Standalone Application (v 2.0.1)
2. SSO CAS plus LDAP ADS against Alfresco Compliant Standalone Application
3. SSO CAS plus LDAP ADS against Liferay Portal (4.3) and Alfresco Portlet (v 2.0.1)





Summary

A generic solution of SSO CAS plus LDAP
ADS against

- Alfresco Standalone Application (v 2.0.1)
- Alfresco Compliant Standalone Application
- Liferay Portal (4.3) and Alfresco Portlet (v 2.1)

Instructions for **Do-It-Yourselfers**

Sources - <http://liferay.cignex.com>

Q & A

